Supplementary Material: Object Discovery from Spike Timing

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1 Experiments, continued

1.1 Are the baselines handicapped?

Two objections to Table 1 of the main paper deserve measurements rather than argument: that the baselines are evaluated away from their published setting, and that the matched budget starves them.

Released weights. Only Slot Attention ships trained weights; the AKOrN, CAE and SynCx repositories release data but no checkpoints. Evaluated directly, at roughly $320\times$ our exposure budget, its released model scores $.3747/.2219/.1165$ on CLEVR, $.4742/.2381/.2355$ on Tetrominoes and $.3205/.0980/.1204$ on Multi-dSprites. We remain ahead in all nine cells, by 0.377 to 0.720, so the comparison does not rest on our retraining of it. The extra data also helps it far more *off* its training distribution than on it — worth $+.0875$ FG-ARI on CLEVR but $+.3003$ on Tetrominoes and $+.1275$ on Multi-dSprites — so a data-efficiency conclusion drawn from an in-domain column alone would not transfer.

Budget. Comparing each baseline against its own published number, by its own evaluation code, isolates the effect of the shortened schedule. Two of the six are *not* budget-limited: AKOrN (conv) reaches 60.5 at $1/210$ of its schedule against a published 59.0 ± 4.3, and CAE reaches 10.3 at $1/64$ against a published 4.0. For those two the margin in Table 1 of the main paper is not an artefact of a short schedule. The other four are limited — AKOrN (attn) 60.4 against 91.0 ± 0.5, ItrSA 58.1 against 82.5 ± 1.7, CAE++ 16.7 against 27.0, and CtCAE 12.0 against 54.0 — and against those the result should be read as a data-efficiency claim. We state both readings rather than only the favourable one. Full figures are in the supplementary document.

1.2 Rollout length

The readout integrates spiking activity over T simulation steps and reads the second half of the window. Table 1 varies T with the model fixed.

T	FG-ARI	FG-IOU	OBJ-IOU	Δ FG-ARI
1024	.8421	.7545	.6410	—
512	.8402	**.7594**	.6340	$-.0019$
256	.8159	.7358	.6084	$-.0262$
128	.5796	.4988	.3985	$-.2625$

Table 1: CLEVR, model fixed, readout window $T/2$. Halving the rollout is free; a quarter costs 0.02–0.03; an eighth fails. This table and the scaling figures below are reported *without* the minimum-region-size rule, because both are within-model comparisons at a fixed readout and the rule would shift every row by the same small amount.

Halving the rollout costs 0.0019 FG-ARI — below any difference we can resolve at one seed — and *improves* foreground IoU. Performance degrades gently to $T = 256$ and collapses at $T = 128$, where the model also under-segments, predicting a foreground fraction of 0.1148 against a truth of 0.2115. We therefore use $T = 512$ as the default operating point and $T = 1024$ for the reported numbers. Shorter windows also fragment more: mean predicted regions per image rise from 8.0 at $T = 1024$ to 13.3 at $T = 256$, against 6.36 true objects.

Scaling with training data. Continuing training on additional unseen CLEVR images, in steps of $1{,}000$, separates the three metrics. FG-ARI rises from $.8421$ at $10{,}000$ images to $.8469$ at $14{,}000$ and then falls to $.8459$ at $15{,}000$; its step-to-step changes are $+.0005$, $+.0023$, $+.0011$, $+.0009$, $-.0010$. Foreground IoU and matched-object IoU do not turn over the same range, gaining $+.0157$ and $+.0124$ and still rising at $15{,}000$. More in-domain data therefore stops improving *how* patches are grouped well before it stops improving *where the boundaries fall*. Mean predicted regions hold at 7.7–8.0 throughout, against 6.36 true objects, so the extra data sharpens the partition without changing how much it splits. A single 0.0010 decrease at one seed is not proof of saturation, and we report it as the shape of the curve rather than a converged value.

In-domain data is not the most useful data. The same $5{,}000$ additional images buy more than twice as much when they come from a different dataset. Continuing on CLEVR raises CLEVR FG-ARI by $+.0038$; continuing on Tetrominoes

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instead raises it by $+.0084$, a $2.2\times$ larger gain at an identical budget. Together with the transfer result in Table 1 of the main paper, this argues that the model is not fitting a domain but learning a grouping rule that the extra renders refine.

2 Supplementary Tables

2.1 Released weights

These figures are given in prose in the main paper.

2.2 Training budget per baseline

2.3 Scaling with training data

	CLEVR	Tetrominoes	Multi-dSprites
Slot Attention, matched budget	.2872 / .2220 / .0801	.1739 / .2192 / .1071	.1930 / .0876 / .0680
Slot Attention, released weights	.3747 / .2219 / .1165	.4742 / .2381 / .2355	.3205 / .0980 / .1204
Ours, matched budget	**.8421 / .7545 / .6410**	**.8580 / .7885 / .8173**	**.6974 / .8177 / .7628**

Table 2: Released weights ($\approx$32M exposures) against the matched-budget retraining ($100{,}032$ exposures) and against our model. Triples are ARI / F-IOU / O-IOU. All three rows are reported *without* the minimum-region-size rule, so the released and retrained Slot Attention rows are compared on identical terms; Table 1 of the main paper applies the rule to every model and our row there reads $.8408/.7528/.6387$, $.8568/.7950/.8170$ and $.6966/.8190/.7625$.

Model	Budget	Ours	Published
AKOrN (conv)	$1/210$	**60.5**	59.0 ± 4.3
CAE	$1/64$	**10.3**	4.0
AKOrN (attn)	$1/210$	60.4	91.0 ± 0.5
ItrSA	$1/210$	58.1	82.5 ± 1.7
CAE++	$1/64$	16.7	27.0
CtCAE	$1/64$	12.0	54.0

Table 3: Each baseline at our budget against its own published figure, by its own metric and evaluation code. *Budget* is the fraction of the published training schedule our matched-exposure protocol gives it. The top block is *not* budget-limited: AKOrN (conv) exceeds its published CLEVR number at $1/210$ of its schedule and CAE exceeds its own at $1/64$. The bottom block is.

Images	FG-ARI	FG-IOU	OBJ-IOU	Δ FG-ARI
10,000	.8421	.7545	.6410	—
11,000	.8426	.7539	.6427	$+.0005$
12,000	**.8449**	**.7607**	**.6461**	$+.0028$

Table 4: *Images* is the cumulative count of unique training images. Each row adds $1{,}000$ previously unseen CLEVR images for 10 epochs, resuming from the previous row. CLEVR evaluation.